

6 Arrays and strings

An Array consists of data items of the same type, all with the same name. The size of an Array is “static”. Once you have created an “array”, you cannot change the number of elements it holds. If you create an array with 42 occurrences, then it will always have 42 elements. An array is a Data Structure. If you wish to create an array where the number of elements can be changed, you use class Vector. In Java, an array is a group of contiguous memory locations that all have the same name and same type.

This declares that x is an array of char variables.

```
char[] x;
```

Allocating the array decides the number of elements.

```
x = new char[5];
```

If you remember “x” is a reference, the syntax above is more understandable.

When you allocate an array, the elements are automatically initialized.

Primitives:

numeric primitives are zeroed,

char primitives are made spaces,

boolean primitives are made false

References:

For an array of any other type the “references” are made null.

```
x[0] = 'Z';
```

This assigns the char 'Z' to the first array position.

```
x[1] = 'q';
```

This assigns the char 'q' to the second array position.

An array is a full-fledge object. You can declare an array either of two ways:

```
char[] x; or char [];
```

If you put the brackets on the data type, you can declare multiple references as all being the same type of array,

```
char[] x, y, z; In this case, x, y and z are all declared as char arrays.
```

To learn the number of elements in the array, you call a property called “length.” For example:

```
int y = 0;
```

```
int[] x;
```

```
x = new int[5];
```

```
y = x.length;
```